

E-Commerce Orders Data Analytics

September 29, 2026

Overview :

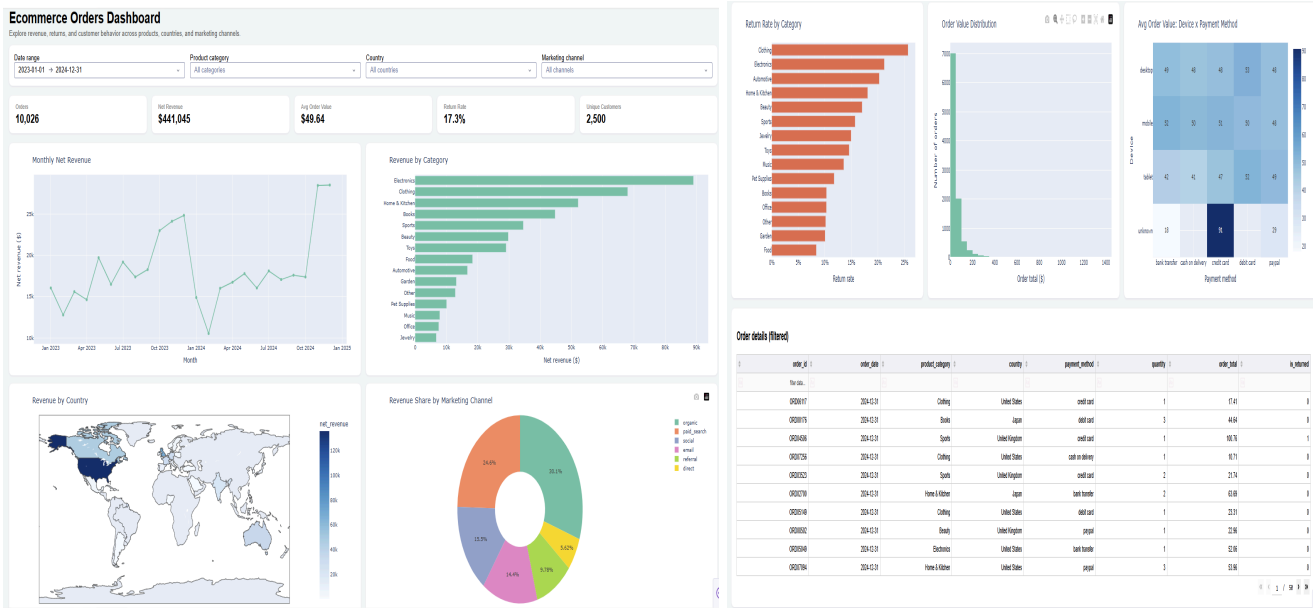
The objective of this project is to clean and analyze e-commerce order data to identify meaningful patterns in sales, revenue, returns, customers, payment methods and present the findings through an interactive dashboard.

DataSet :

- **Source:** Kaggle – E-Commerce Orders
- **Published by:** Mohammed Slimane
- **Dataset:** <https://www.kaggle.com/datasets/mohammedslimane/ecommerce-orders>
- **Data includes:** Order, customer, product, revenue, returns, payment, device, country, and marketing-channel information.
- **Analysis focus:** Sales performance, customer behaviour and returns.
- **Tools:** Python, Pandas, Matplotlib, Seaborn, Plotly and Dash.

Project Workflow :

Raw Data → Data Cleaning → Data Wrangling → Exploratory Data Analysis → Visualization → Interactive Dashboard



Key insights:

The analysis focused on identifying patterns in **revenue, customer behaviour, orders and returns**.

- **Revenue:** 2024 was weaker than 2023 for most of the year but recovered strongly in November and December, with revenue exceeding **\$28K+ per month**. (Seasonal or Promotional effect during the final two months.).
- **Customer behaviour:** Weekend revenue was stronger, with **Sunday** generating the highest net revenue and Wednesday the lowest .
- **Customers:** Returning customers represented **75% of customer volume** and generated over **\$300K in revenue**, compared with approximately \$110K from new customers.
- **Customer demographics:** **35–44** years is the **largest age segment** (26%), followed by 45–54 (21.4%) and 25–34 (20.4%); overall volume is concentrated among ages 25–54.
- **Return Rate :** **18–24** years has the **lowest revenue** but the **highest return rate** (~25%), suggesting an opportunity to improve younger-customer acquisition and retention.
- **Products:** **Electronics** generated the **highest revenue**, while **Clothing** had the **highest return rate**. Electronics and Clothing therefore require further investigation because of their combination of revenue and returns.
- **Payment Methods:** **Credit card** is the largest contributor to net revenue, followed by **debit card**.
- **Geography:** The **United States generated the highest revenue**, followed by the UK and Canada. High return rates in smaller markets did not necessarily represent the greatest financial impact.
- **Channels and devices:** Organic search was the largest marketing channel, while **mobile and desktop** accounted for nearly **90% of sales volume**.

Challenges Faced:

- Cleaning duplicate records and handling missing values.
- Validating dates, numerical fields and values, discounts, and categorical data.
- Interpreting multiple variables together rather than relying on a single metric.
- Converting multiple findings into a clear and interactive dashboard.

What We Learned:

- **Data cleaning is essential** before performing reliable analysis.
- A single metric can be misleading; **multiple business variables should be analyzed together**.
- The **median** can provide a more useful benchmark than the mean when data contains extreme values.
- Effective data analytics requires both **technical skills and clear communication**.
- The project gave us practical experience moving from **Raw Data → Cleaning → EDA → Visualization → Interactive Dashboard**.